MA1220 – Calculus II

Date: October 14, 2023 Maximum Marks: 50

Time: 120 minutes, 09:30 AM – 11:30 AM

Q.1 Answer the following. (No justification required.) (8 marks)

(a) If $f(x) \geq g(x)$ on the interval $[a, \infty)$, then if $\int_a^\infty f(x)dx$ converges, then $\int_a^\infty g(x)dx$ also converges.

(b) $\partial^2 f/\partial x \partial y = \partial^2 f/\partial y \partial x$ is always true.

(c) If $f_x = f_y = 0$ at a point (x, y), then f is differentiable at (x, y).

(d) $D_v(\partial f/\partial x_2)(-3, 4) = \partial^2 f/\partial x_1 \partial x_2(-3, 4)$ if $v = \langle 1, 0 \rangle$.

(e) If $f(1) = 12$, f' is continuous, and $\int_1^4 f'(x)\,dx = 17$, then $f(4) = ?$

(f) Is $\int_{-1}^1 (1/x)\,dx$ convergent or divergent?

(g) Describe the domain D of the function $f(x, y) = \sqrt{(1 - x^2 - y^2)} \cdot \ln(x^2 + y^2 - 1)$.

(h) Is it true that $\int_{-\infty}^\infty x\,dx = \lim_{t\to\infty} \int_{-t}^t x\,dx$?

Q.2 Answer the following:

(a) Find the directional derivative of the function $f(u, v) = u^2 e^{-v}$ at the point $(3, 0)$ in the direction of the vector $3\mathbf{i} + 4\mathbf{j}$. (3 marks)

(b) By comparing (without integrating), determine whether the integral $\int_1^\infty 1/\sqrt{(x^3 + 1)}\,dx$ converges or diverges. (2 marks)

(c) Find the local minima, maxima, and saddle points of the function $f(x, y) = x^2 + xy + y^2 + y$ in $\mathbb{R}^2$. (3 marks)

Q.3 Answer the following:

(a) Find the volume V obtained by rotating the region enclosed by the graphs of the functions $f(x) = x(5 - x)$ and $g(x) = 8 - x(5 - x)$ about the y-axis. (4 marks)

(b) Find the area of the region bounded by the graphs of $y = 8/x^2$, $y = 8x$, and $y = x$. (4 marks)

Q.4 Answer the following:

(a) If the region $R = \{(x, y) \mid x \geq 1, 0 \leq y \leq 1/x\}$ is rotated about the x-axis, by using improper integrals show that:

(i) The volume of the resulting solid is finite, and find the volume. (2 marks)

(ii) The surface area is infinite. (3 marks)

(b) If $f(x, y) = x^{2023}y / (x^{4046} + y^2)$, does $\lim_{(x,y)\to(0,0)} f(x, y)$ exist? (3 marks)

Q.5 Answer the following:

(a) Determine the function f(x) and all values of a such that $\int_a^x f(t)\,dt = x^2 + x - 6$. (2 marks)

(b) Evaluate $g''(\pi/6)$, where $g(y) = \int_3^y f(x)\,dx$ and $f(x) = \int_0^{\sin x} \sqrt{(1 + t^2)}\,dt$. (2 marks)

(c) Using Riemann integration, find the value of $\lim_{n\to\infty} \Sigma_{k=1}^n\, n/(n^2 + k^2)$. (2 marks)

(d) Does there exist a function $f(x, y)$ such that $f_x(x, y) = xy$ and $f_y(x, y) = x^2$? If yes, find f. If no, explain your reason. (2 marks)

Q.6 Answer the following:

(a) Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x, y) = \sigma x + \pi y$ for all $(x, y) \in \mathbb{R}^2$, where σ and π are constants. Prove (using the ε–δ definition) that f is continuous on $\mathbb{R}^2$. (4 marks)

(b) Let $g : \mathbb{R}^2 \to \mathbb{R}$ be defined by $g(x, y) = 5x + 4y^2$. Prove (using the steps given below) that g is differentiable at $(2, 1)$. (4 marks)

Step 1. Compute the linearization $L(x, y)$ (the equation of the tangent plane) of $g(x, y)$ at $(2, 1)$.

Step 2. Compute the difference $g(x, y) - L(x, y)$.

Step 3. Show that for all $(x, y) \neq (2, 1)$, one can write $0 \leq |g(x, y) - L(x, y)| / \sqrt{((x - 2)^2 + (y - 1)^2)} \leq A|y - 1|$ for some constant A. Find one such A.

Step 4. Apply the definition of differentiability for $g(x, y)$ at $(2, 1)$.